

ECO102: Introduction to Macroeconomics

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SECTION: 05

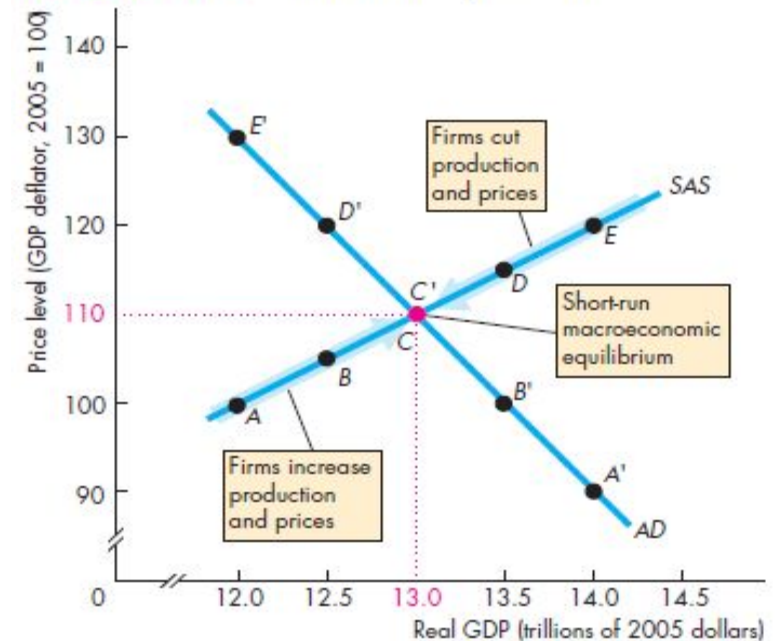
LECTURE 11-12

CHAPTER 27: SHORT RUN & LONG RUN EQUILIBRIUM

Short Run Macroeconomic Equilibrium

- ▶ **Short-run macroeconomic equilibrium** occurs when the quantity of real GDP demanded equals the quantity of real GDP supplied.
- ▶ That is, short-run macroeconomic equilibrium occurs at the point of intersection of the AD curve and the SAS curve.
- ▶ In the short run, the money wage rate is fixed. It does not adjust to move the economy to full employment. So in the short run, real GDP can be greater than or less than potential GDP.

FIGURE 27.6 Short-Run Equilibrium

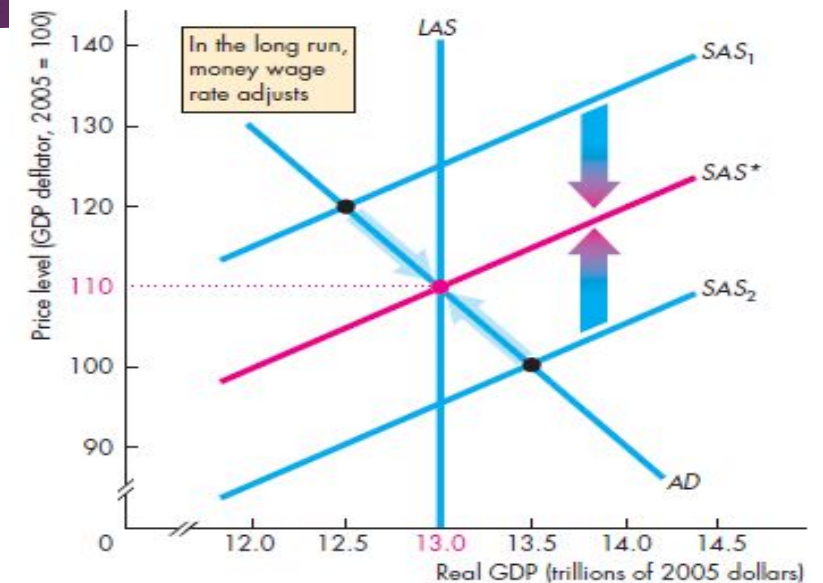


Short-run macroeconomic equilibrium occurs when real GDP demanded equals real GDP supplied—at the intersection of the aggregate demand curve (AD) and the short-run aggregate supply curve (SAS).

Long Run Macroeconomic Equilibrium

- ▶ **Long-run macroeconomic equilibrium** occurs when real GDP equals potential GDP—equivalently, when the economy is on its *LAS* curve.
- ▶ When the economy is away from long-run equilibrium, the money wage rate adjusts.
- ▶ If the money wage rate is too high, short-run equilibrium is below potential GDP and the unemployment rate is above the natural rate. With an excess supply of labor, the money wage rate falls.

FIGURE 27.7 Long-Run Equilibrium

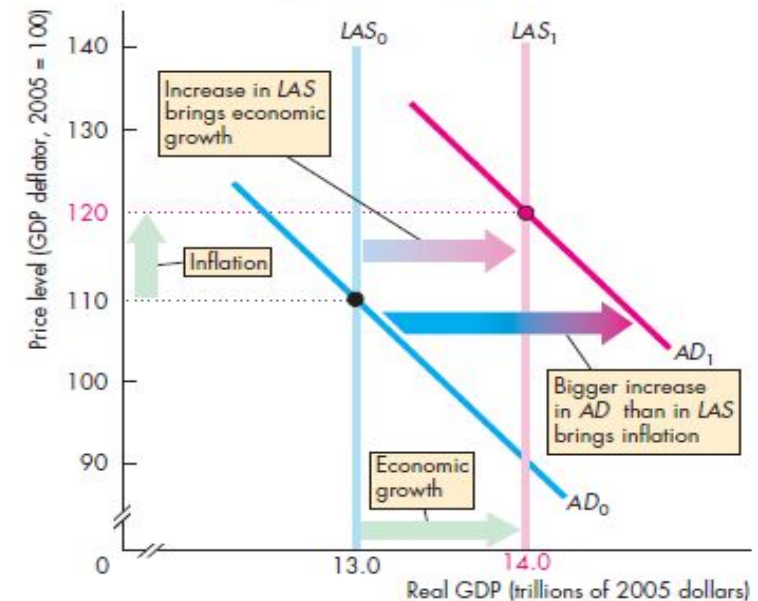


In long-run macroeconomic equilibrium, real GDP equals potential GDP. So long-run equilibrium occurs where the aggregate demand curve, *AD*, intersects the long-run aggregate supply curve, *LAS*. In the long run, aggregate demand determines the price level and has no effect on real GDP. The money wage rate adjusts in the long run, so that the *SAS* curve intersects the *LAS* curve at the long-run equilibrium price level.

Economic Growth and Inflation in the AS-AD Model

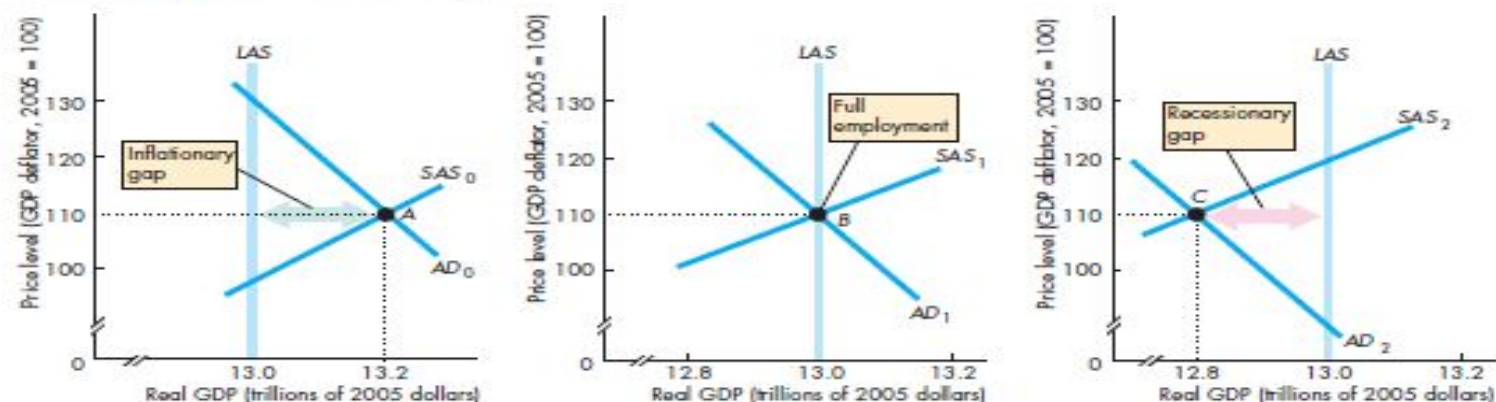
- ▶ Economic growth results from a growing labor force and increasing labor productivity, which together make potential GDP grow.
- ▶ Inflation results from a growing quantity of money that outpaces the growth of potential GDP.
- ▶ The AS-AD model explains economic growth as increasing long-run aggregate supply and it explains inflation as a persistent increase in aggregate demand at a faster pace than that of the increase in potential GDP.
- ▶ If aggregate demand were to increase at the same pace as long-run aggregate supply, real GDP would grow with no inflation.

FIGURE 27.8 Economic Growth and Inflation



The Business Cycle in the AS-AD Model

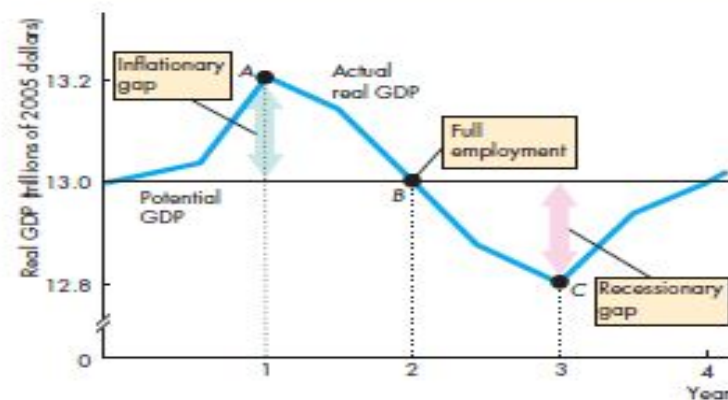
FIGURE 27.9 The Business Cycle



(a) Above full-employment equilibrium

(b) Full-employment equilibrium

(c) Below full-employment equilibrium



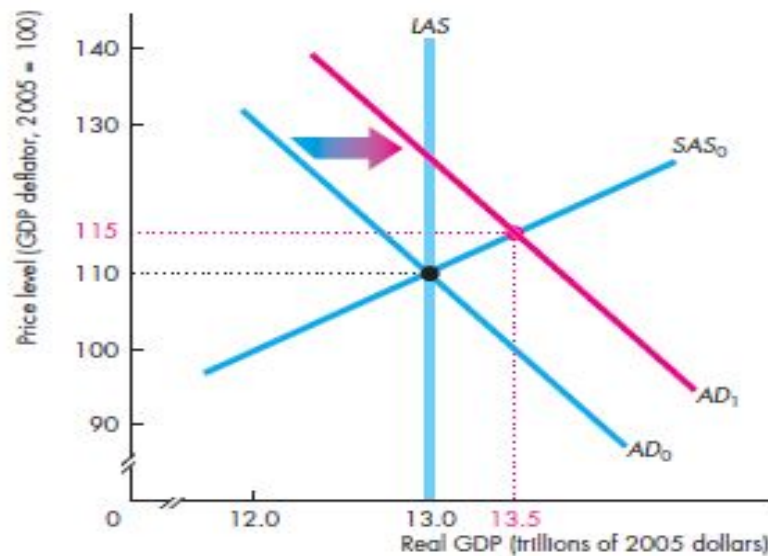
(d) Fluctuations in real GDP

Part (a) shows an above full-employment equilibrium in year 1; part (b) shows a full-employment equilibrium in year 2; and part (c) shows a below full-employment equilibrium in year 3. Part (d) shows how real GDP fluctuates around potential GDP in a business cycle.

In year 1, an inflationary gap exists and the economy is at point A in parts (a) and (d). In year 2, the economy is at full employment and the economy is at point B in parts (b) and (d). In year 3, a recessionary gap exists and the economy is at point C in parts (c) and (d).

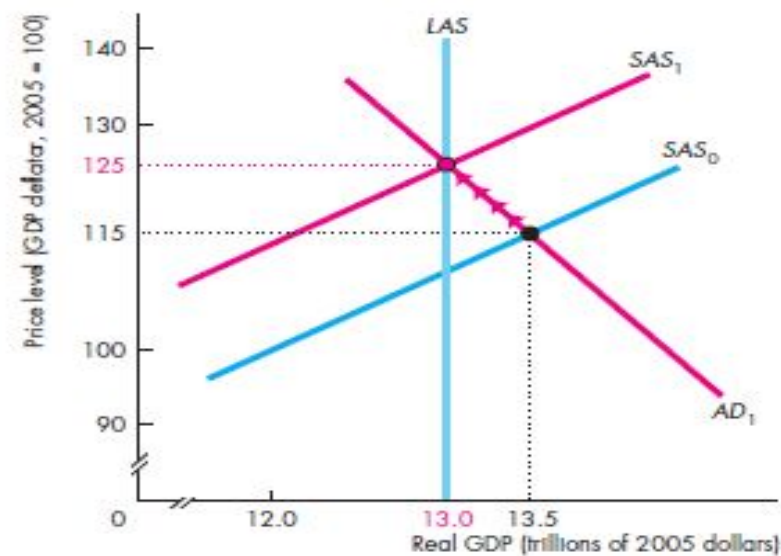
Fluctuations in Aggregate Demand

FIGURE 27.10 An Increase in Aggregate Demand



(a) Short-run effect

An increase in aggregate demand shifts the aggregate demand curve from AD_0 to AD_1 . In short-run equilibrium, real GDP increases to \$13.5 trillion and the price level rises to 115. In this situation, an inflationary gap exists. In the long run in part (b), the money wage rate rises and the short-run aggregate



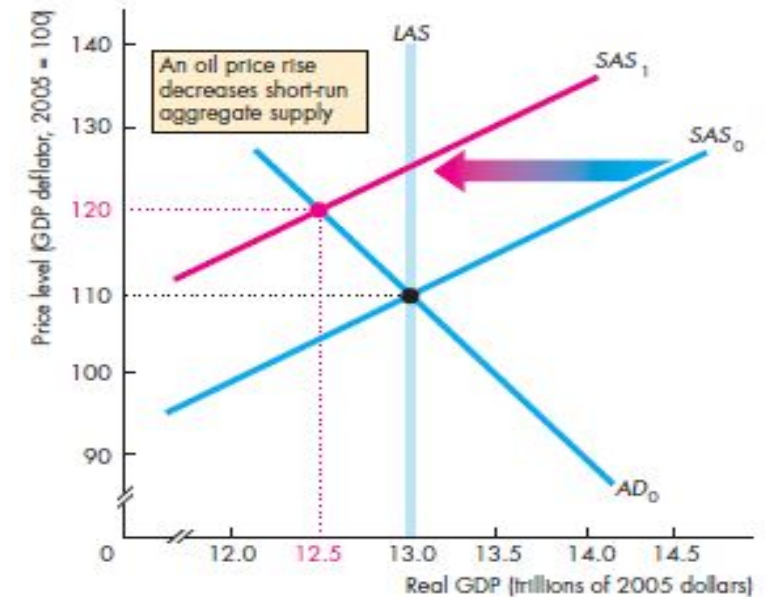
(b) Long-run effect

gate supply curve shifts leftward. As short-run aggregate supply decreases, the SAS curve shifts from SAS_0 to SAS_1 and intersects the aggregate demand curve AD_1 at higher price levels and real GDP decreases. Eventually, the price level rises to 125 and real GDP decreases to \$13 trillion—potential GDP.

Fluctuations in Aggregate Supply

- ▶ Fluctuations in short-run aggregate supply can bring fluctuations in real GDP around potential GDP.
- ▶ Suppose that initially real GDP equals potential GDP. Then there is a large but temporary rise in the price of oil. What happens to real GDP and the price level?
- ▶ Faced with higher energy and transportation costs, firms decrease production. Short-run aggregate supply decreases, and the short-run aggregate supply curve shifts leftward to SAS_1 .
- ▶ Because real GDP decreases, the economy experiences recession. Because the price level increases, the economy experiences inflation. A combination of recession and inflation, called **stagflation**.
- ▶ When the price of oil returns to its original level, the economy returns to full employment.

FIGURE 27.11 A Decrease in Aggregate Supply



An increase in the price of oil decreases short-run aggregate supply and shifts the short-run aggregate supply curve from SAS_0 to SAS_1 . Real GDP falls from \$13 trillion to \$12.5 trillion, and the price level rises from 110 to 120. The economy experiences stagflation.

Macroeconomic School of Thought (SELF-STUDY)

- ▶ The Classical View
- ▶ The Keynesian View
- ▶ The Monetarist View

- ▶ **READ THIS BY YOURSELF FROM THE BOOK**